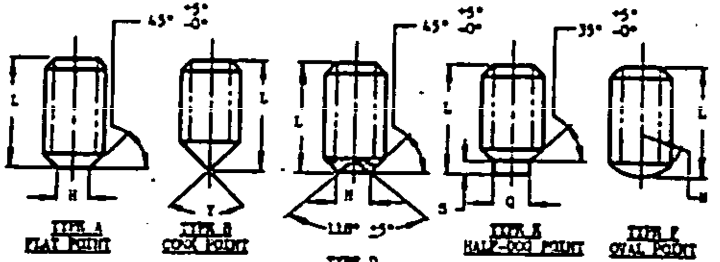
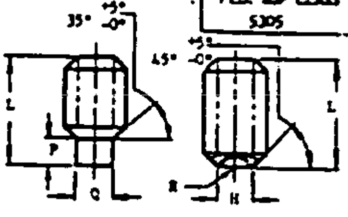
Proj. No. 5305-1294

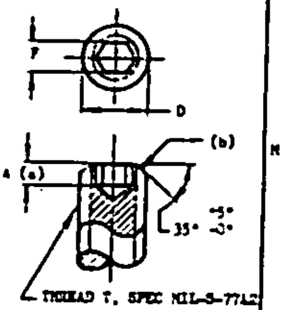
FED. SUP CLASS 5305



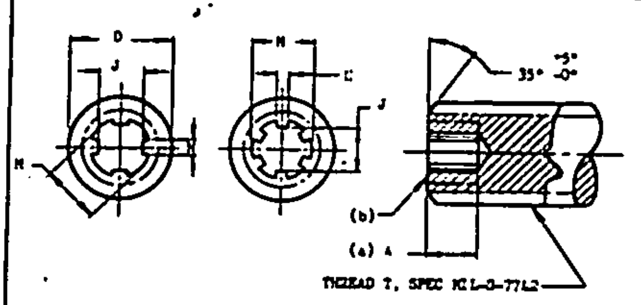
PREFERRED POINT STYLES



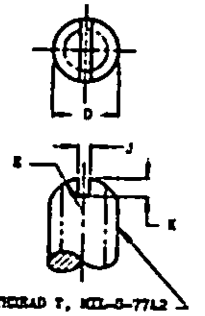
INACTIVE FOR DESIGN



HEX SOCKET HEAD



FLUTED SOCKET HEAD
(NO. 2, 4, AND 6 SIZES ONLY)



SLOTTED HEAD
INACTIVE FOR DESIGN

NOMINAL SIZE	THREAD T	D MAX DIA	FLUTED SOCKET HEAD						
			NUMBER OF FLUTES	J SOCKET DIAMETER, MINOR		M SOCKET DIAMETER, MAJOR		N SOCKET LAND WIDTH	
				MAX	MIN	MAX	MIN	MAX	MIN
NO. 2	NO. 2-56UNC-3A	.086	4	.038	.0375	.050	.049	.017	.016
NO. 4	NO. 4-40UNC-3A	.112	6	.051	.050	.062	.061	.014	.013
NO. 6	NO. 6-32UNC-3A	.138	4	.056	.055	.079	.078	.023	.022

NOMINAL SIZE	THREAD T	D MAX DIA	HEX SOCK HEAD		SLOTTED HEAD			TYPES A AND D			TYPE B		TYPES E AND C				TYPE F
			F		E RAD	J NOM	K NOM	H DIA		R SPHER RAD	Y 118° ±2° FOR LENGTHS AND UNDER	Y 90° ±2° FOR LENGTHS AND OVER	Q		S	P	U RADIUS ±.016
			MAX	MIN				MAX	MIN				MAX	MIN			
NO. 2 NO. 3 NO. 4	NO. 2-56UNC-3A NO. 3-48UNC-3A NO. 4-40UNC-3A	.086 .099 .112	.0355 .0510 .0510	.0350 .0500 .0500	.057 .064 .067	.018 .018 .020	.022 .025 .028	.047 .054 .061	.039 .045 .051	.031 .037 .042	3/32 7/64 1/8	7/64 1/8 5/32	.057 .066 .075	.053 .062 .070	.022 .025 .028	.043 .052 .056	.063 .078 .094
NO. 5 NO. 6 NO. 8	NO. 5-40UNC-3A NO. 6-32UNC-3A NO. 8-32UNC-3A	.125 .138 .164	.0635 .0635 .0791	.0625 .0625 .0781	.075 .083 .099	.023 .025 .029	.031 .035 .041	.067 .074 .087	.057 .064 .076	.045 .045 .050	1/8 1/8 3/16	3/16 3/16 1/4	.083 .092 .109	.078 .087 .103	.030 .035 .040	.060 .070 .080	.094 .109 .125
NO. 10 NO. 10 1/4	NO. 10-24UNC-3A NO. 10-32UNF-3A 1/4-20UNC-3A	.190 .190 .250	.0947 .0947 .1270	.0937 .0937 .1250	.112 .112 .151	.032 .032 .045	.048 .048 .063	.102 .102 .132	.088 .088 .118	.060 .060	3/16 3/16 1/4	1/4 1/4 5/16	.127 .127 .156	.120 .120 .149	.045 .045 .063	.090 .090 .125	.141 .141 .188
1/4 5/16 5/16	1/4-28UNF-3A 5/16-18UNC-3A 5/16-24UNF-3A	.250 .3125 .3125	.1270 .1582 .1582	.1250 .1562 .1562	.151 .188 .188	.045 .051 .051	.063 .078 .078	.132 .172 .172	.118 .156 .156		1/4 5/16 5/16	5/16 3/8 3/8	.156 .203 .203	.149 .195 .195	.063 .078 .078	.125 .156 .156	.188 .234 .234
3/8 3/8 1/2	3/8-16UNC-3A 3/8-24UNF-3A 1/2-13UNC-3A	.375 .375 .500	.1895 .1895 .2520	.1875 .1875 .2500	.226 .226 .301	.064 .064 .081	.094 .094 .125	.212 .212 .291	.194 .194 .270		3/8 3/8 1/2	7/16 7/16 9/16	.250 .250 .344	.241 .241 .334	.094 .094 .125	.188 .188 .250	.281 .281 .375
1/2	1/2-20UNF-3A	.500	.2520	.2500	.301	.081	.125	.291	.270		1/2	9/16	.344	.334	.125	.250	.375

⑤ FOR CHANGES SEE SHEET 2

APPROVED 27 Dec 54 REVISED ① 26 Feb 63 ② 24 Jan 64 ③ 29 DEC 72 ④ 8 NOV 74 ⑤ 23 JUN 1975

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all noncorrelated standard drawings for the same product and shall become effective for the procurement of nonstandard supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

PROCUREMENT
SPECIFICATION ⑤
FF-S-200

AIR FORCE-NAVY AERONAUTICAL STANDARD

SETSCREW-HEXAGON AND FLUTED SOCKET, HEADLESS

AN565

SUPERSEDES:
SUPERSEDES USAF DRAWINGS 565 AND 566

SHEET 1 OF 3

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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THE NO. 2 SIZE IS NON-PREFERRED FOR AIRFRAME USE

NO. 3-48, NO. 5-40, NO. 10-24, 1/4-20, 5/16-18, 3/8-16, AND 1/2-13 COARSE THREAD SETSCREWS INACTIVE FOR DESIGN AFTER 1 JULY 1954

COARSE THREAD

L LENGTH		NO. 2-56	NO. 3-48	NO. 4-40	NO. 5-40	NO. 6-32	NO. 8-32	NO. 10-24	1/4-20	5/16-18	3/8-16	1/2-13
1/16	±.010	2-1		4-1								
1/8	±.010	2-2	3-2	4-2		6-2	8-2					
3/16	±.010	2-3	3-3	4-3	5-3	6-3	8-3	10-3	416-3			
1/4	±.010	2-4	3-4	4-4	5-4	6-4	8-4	10-4	416-4	516-4		
5/16	±.010	2-5	3-5	4-5	5-5	6-5	8-5	10-5	416-5	516-5		
3/8	±.010			4-6	5-6	6-6	8-6	10-6	416-6	516-6	616-6	
7/16	±.010			4-7	5-7	6-7	8-7	10-7				
1/2	±.010			4-8		6-8	8-8	10-8	416-8	516-8	616-8	816-8
9/16	±.010			4-9		6-9	8-9	10-9				
5/8	±.010			4-10		6-10	8-10	10-10	416-10	516-10	616-10	816-10
3/4	±.011					6-12	8-12	10-12	416-12	516-12	616-12	816-12
7/8	±.013					6-14	8-14		416-14	516-14	616-14	816-14
1	±.015						8-16		416-16	516-16	616-16	816-16
1-1/4	±.019								416-20	516-20	616-20	816-20
1-1/2	±.022								416-24	516-24	616-24	816-24
1-3/4	±.026									516-28	616-28	816-28
2	±.030									516-32	616-32	816-32
2-1/4	±.031											816-36
2-1/2	±.031											816-40
2-3/4	±.031											816-44
3	±.031											816-48

FINE THREAD

L LENGTH		DASH NUMBERS				
		NO. 10-32	1/4-28	5/16-24	3/8-24	1/2-20
1/8	±.010	1032-2				
3/16	±.010	1032-3	428-3			
1/4	±.010	1032-4	428-4	524-4		
5/16	±.010	1032-5	428-5	524-5		
3/8	±.010	1032-6	428-6	524-6	624-6	820-6
7/16	±.010	1032-7	428-7			
1/2	±.010	1032-8	428-8	524-8	624-8	820-8
9/16	±.010	1032-9	428-9			
5/8	±.010	1032-10	428-10	524-10	624-10	820-10
3/4	±.011	1032-12	428-12	524-12	624-12	820-12
7/8	±.013		428-14	524-14	624-14	820-14
1	±.015	1032-16	428-16	524-16	624-16	820-16
1-1/4	±.019	1032-20	428-20	524-20	624-20	820-20
1-1/2	±.022		428-24	524-24	624-24	820-24
1-3/4	±.026			524-28	624-28	820-28
2	±.030			524-32	624-32	820-32
2-1/4	±.031					
2-1/2	±.031					820-40
2-3/4	±.031					
3	±.031					820-48

⑤

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NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

APPROVED 26 Feb 51 REVISED ⑤ [illegible] 2

PROCUREMENT SPECIFICATION ⑤
FF-S-200

AIR FORCE-NAVY AERONAUTICAL STANDARD

SETSCREW-HEXAGON AND FLUTED SOCKET, HEADLESS

AN565

SUPERSEDES:
SUPERSEDES USAF DRAWINGS 505 and 556

SHEET 2 OF 3

(a) SOCKET DEPTH A: THE DEPTH OF SOCKET SHALL BE AS GREAT AS PRACTICABLE BUT VARYING CONDITIONS RENDER IT INADVISABLE TO SPECIFY DEFINITE VALUES FOR THIS DIMENSION.

(b) SOCKET END OF SCREW SHALL BE FLAT AND CHAMFERED. THE CHAMFER SHALL EXTEND TO THE BOTTOM OF THE THREAD AND THE EDGE BETWEEN FLAT AND CHAMFER SHALL BE SLIGHTLY ROUNDED.

(c) NO. 3-48, 5-40, 10-24, 1/4-20, 5/16-18, 3/8-16, 1/2-13 SETSCREWS INACTIVE FOR DESIGN AFTER 1 JULY 1954. SLOTTED HEAD SETSCREWS INACTIVE FOR DESIGN AFTER 1 JULY 1954. USE HEXAGON SOCKET SETSCREWS OR, FOR NO. 2, 4, AND 6 SIZES, USE EITHER FLUTED OR HEXAGON SOCKET SETSCREWS.

SCREWS WITH NATIONAL CLASS 3 THREAD MAY BE SUBSTITUTED FOR UNIFIED CLASS 3A THREADS UNTIL EXISTING STOCKS OF NATIONAL CLASS 3 SCREWS ARE DEPLETED. UNIFIED CLASS 3A WILL BE MANDATORY ON ALL SCREWS SUPPLIED AFTER 1 JANUARY 1956.

MATERIAL: STEEL, HIGH-GRADE ALLOY (4037, 4137, 8630, 8740) HARDENED BY QUENCHING IN OIL FROM THE HARDENING TEMPERATURE, AND TEMPERED TO A ROCKWELL HARDNESS OF C45-C53, OR EQUIVALENT CRES, MIL-S-7720, COMP 303S OR 303Se.

FINISH: CADMIUM PLATE, SPECIFICATION QQ-P-416, TYPE II, CLASS 2. PARTS WITH CLASS 3 PLATING MAY BE FURNISHED FROM SUPPLIER'S STOCK UNTIL 1 JANUARY 1975.
PASSIVATE, CORROSION RESISTANT STEEL, PER QQ-P-35..

HEXAGON SOCKET AND FLUTED SOCKET SETSCREWS MAY BE SPECIFIED WITH ANY POINT STYLE SHOWN: KNURLED POINT ON TYPE D SETSCREW MAY BE SUPPLIED AS OPTIONAL.

ADD POINT STYLE TYPE BEFORE FIRST DASH NUMBER.

ADD H BETWEEN DASH NUMBERS FOR SETSCREW WITH HEXAGON SOCKET HEAD.

ADD L BETWEEN DASH NUMBERS FOR SETSCREW WITH FLUTED SOCKET HEAD. (NO. 2, 4 AND 6 SIZES, ONLY.)

(c) DASH BETWEEN DASH NUMBERS DESIGNATES SLOTTED HEAD SETSCREW (INACTIVE FOR DESIGN).

ADD C AFTER POINT STYLE TYPE TO DESIGNATE CORROSION RESISTING STEEL SETSCREWS.

EXAMPLES OF PART NOS.:

AN565B1032H12 - SETSCREW WITH TYPE B POINT, WITH HEXAGON SOCKET HEAD, NO. 10-32 SIZE (.190 INCH DIA), 3/4(12/16) IN. LONG, STL.

AN565B4L5 - SETSCREW WITH TYPE B POINT, WITH FLUTED SOCKET HEAD, NO. 4-40 SIZE (.112 INCH DIA), 5/16 INCH LONG, STEEL.

(c) AN565A10-12 - SETSCREW WITH TYPE A POINT, WITH SLOTTED HEAD, NO. 10-24 SIZE (.190 INCH DIA), 3/4(12/16) INCH LONG, STEEL. (INACTIVE FOR DESIGN).

AN565BC1032H12 - SETSCREW WITH TYPE B POINT, WITH HEXAGON SOCKET HEAD, NO. 10-32 SIZE, 3/4 INCH LONG, CORROSION RESISTING STEEL.

THE LIST OF SETSCREWS THAT ARE CONSIDERED AS STANDARD STOCK ITEMS IS INCLUDED IN SPECIFICATION FF-S-103.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ±1/64, DECIMALS ±.005.

IN CASE OF CONFLICT, THE REQUIREMENTS SPECIFIED HEREON SHALL TAKE PRECEDENCE OVER THE REQUIREMENTS IN SPECIFICATION FF-S-103.

APPROVED 26 FEB 54 REVISED ⑤ FOR CHANGES SEE SHEET 2

PROCUREMENT SPECIFICATION FF-S-200 ⑤	AIR FORCE-NAVY AERONAUTICAL STANDARD SETSCREW-HEXAGON AND FLUTED SOCKET, HEADLESS	AN565
	SUPERSEDES USAF DRAWINGS 565 AND 566	SHEET 3 OF 3

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